

Virtual Disk Image Analysis & Troubleshooting

When testing the Aegis-OS system images using QEMU, it is occasionally necessary to inspect the raw virtual disk to verify structural integrity, particularly if the bootloader fails to execute properly. We utilize **TestDisk** to analyze the partition structure and verify boot sector signatures.

Diagnosing Boot Sector Errors

If the system image fails to boot or is unrecognized, follow these steps to analyze the disk structure:

1. Select Partition Table Type

Launch TestDisk and select the virtual disk (`/dev/sda` in this environment). Because the image is being built from scratch and may not yet have a standard partition table written to it, select **[None] Non partitioned media**.

[image] Selecting the appropriate media type for the raw virtual disk.

2. Initiate Disk Analysis

From the disk menu, select **[Analyse]** to scan the current partition structure and search for any lost or misaligned partitions. Correct disk geometry is critical here, but TestDisk will handle standard 512-byte sector sizes for the virtual image.

[image] Initiating the analysis process on the 407 MB virtual disk.

3. Missing Boot Signature (0xAA55)

During the creation of the bootloader, ensuring standard compatibility (like the Multiboot 1 header) is crucial. A common error during early development is the absence of the Master Boot Record (MBR) boot signature. If TestDisk returns the error `Partition sector doesn't have the endmark 0xAA55`, the binary is missing the standard 55 AA magic number at the end of the boot sector (byte 510 and 511).

[image] TestDisk successfully identifying the missing 0xAA55 boot signature.

4. Search Results & Resolution

If a Quick Search yields **No partition found or selected for recovery**, it confirms the image is entirely raw or the partition table is entirely unwritten.

[image] Final result indicating no recoverable standard partitions.

Next Steps:

- Verify the linker script to ensure the bootloader header and signature are placed at the correct memory offset.

- Recompile the system image to include the necessary 0xAA55 endmark before the next QEMU test run.